

QUESTION 3: (Start a new page)

- i) If the probability of a hit in a single run is 0.1. Calculate the probability of a jet fighter getting exactly 2 hits on a target in 20 runs at the target.
- ii) a) Show that the equation

$$2x^3 - 3x^2 + 0.99 = 0$$
 has a root near $x = 1$.
- b) Attempt to find an improved value of this root by using Newton's Method once, starting with $x_0 = 1$.
- c) Explain why this attempt fails.
- iii) The polynomial $2x^3 + 3x^2 + ax - 6$ has $(x + 3)$ and $(2x + b)$ as factors.
 Find 'a' and 'b'

QUESTION 4: (Start a new page)

- i) By using the principle of mathematical induction prove

$$\sum_{k=1}^n 5^k = 5 \left(\frac{5^n - 1}{4} \right)$$

- ii) a) What is a primitive function of $e^{f(x)} \cdot f'(x)$?

b) Using part (a) evaluate $\int_0^1 \frac{e^{\cos^{-1}x} dx}{\sqrt{1-x^2}}$ (leave answer in irrational form)

- iii) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$

- (a) Find the equation of the tangent l to the parabola at Q .
- (b) Derive the equation of the chord PQ and show that $pq = -1$ when PQ is focal chord.
- (c) Find the acute angle between the tangent l and the chord PQ
 If $p = 3$ and $q = -0.2$.

QUESTION 5 (Start a new page)

- i) The velocity of a particle moving in S.H.M. in a straight line is given by

$$v^2 = 4x - x^2 \text{ ms}^{-1} \text{ where } x \text{ is displacement in } m$$

- a) Find the two points between which the particle is oscillating.
- b) Find the centre of the motion.
- c) Find the maximum speed of the particle.
- d) Find the acceleration of the particle in terms of x .

ii) Prove the identity $\frac{2 \cos A}{\operatorname{cosec} A - 2 \sin A} = \tan 2A$

- iii) A person walking along a straight road observes a tower bearing $045^\circ T$, the angle of elevation being 5° . After travelling a distance of 5000m, the tower bears $315^\circ T$ and the angle of elevation is 8° .

- a) Find the height of the tower (to 0.1m).

- b) Determine the angle which the road makes with a line bearing $090^\circ T$.

QUESTION 6: (Start a new page)

i) a) Sketch the curves $y^2 = x$ and $x^2 = 8y$

b) Show that the area formed between these curves is given by

$$\int_0^4 \left(x^{\frac{1}{2}} - \frac{x^2}{8} \right) dx$$

c) The area is rotated about the x axis, find the volume of the solid of revolution so formed.

ii) Let $f(x) = \sin^{-1} x + \cos^{-1} x$ ($0 \leq x \leq 1$)

Find a) $f'(x)$

$$\int_0^1 f(x) dx$$

iii) Newton's Law of cooling states that the rate at which a body cools is proportional to the excess of its temperature above that of its surroundings.

A sphere at a temperature of 70°C is placed in a container at a temperature of 20°C .

a) Show that $T = 20 + 50 e^{-kt}$ is a solution of the differential equation

$$\frac{dT}{dt} = -K(T - 20) \text{ where } K \text{ is a positive constant.}$$

b) If, after 2 minutes, the temperature of the sphere is 60°C approx.. show that

$$T = 20 + 50 e^{-0.11t}$$

c) Find the temperature of the sphere after 4 minutes (to the nearest degree).

QUESTION 7 (Start a new page)

i) A ball is thrown from a height 1 metre from the ground and is caught, without bouncing, 2 seconds later at a distance of 50m, also at a height of 1m. Assuming no air resistance and that gravity is approx. 10ms^{-2} you may assume $x = Vt \cos \theta$ and $y = -\frac{1}{2}gt^2 + Vt \sin \theta$

Find a) The velocity and angle of projection of the ball.

b) The maximum height of the ball above the ground during its flight.

ii) For the expansion of $(a + bx)^n$ in ascending powers of x

a) Show that the expression for the ratio of the r^{th} and $(r+1)^{\text{th}}$ terms

is $\frac{T_{r+1}}{T_r} = \frac{(n-r+1)}{r} \cdot \frac{bx}{a}$

b) Two consecutive coefficients in the expansion $(3+x)^{15}$ are equal. Find which terms these are.

Q1.

(a). $\frac{2x+1}{x-2} \geq 1$

$\frac{2x+1}{x-2} - 1 \geq 0$

$\frac{2x+1 - (x-2)}{x-2} \geq 0$

$\frac{x+3}{x-2} \geq 0$

$x+3 \geq 0 \cap x-2 \leq 0$
 $x \geq -3 \cap x \leq 2$
 $-3 \leq x \leq 2$



$x \geq -3 \cap x \leq 2$

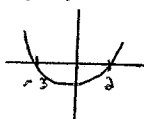
or

$(2x+1)(x-2) \geq (x-2)^2$

$2x^2 - 3x - 2 \geq x^2 - 4x + 4$

$x^2 + x - 6 \geq 0$

$(x+3)(x-2) \geq 0$



$x \geq -3 \cap x \leq 2$

(ii) $\int \frac{\pi \sin x \cos x}{\sqrt{4 - \cos^2 x}}$

Let $u = \cos x$

$\frac{du}{dx} = -\sin x$

when $x = \pi/2$, $u = 0$
 when $x = 0$, $u = 1$

$I = \int_1^0 \frac{-du}{\sqrt{4-u^2}}$

$= \int_0^1 \frac{du}{\sqrt{4-u^2}}$

$= [\sin^{-1}(\frac{u}{2})]_0^1$

$= \sin^{-1}(\frac{1}{2})$

$= \frac{\pi}{6}$

(iii). $(x + \frac{1}{x})^2 - 10(x + \frac{1}{x}) + 24 = 0$

Let $u = (x + \frac{1}{x})$

$(u-4)(u-6) = 0$

$u-4 = 0$ or $u-6 = 0$

$x^2 + 4x + 1 = 0$

$x^2 - 6x + 1 = 0$

$x = \frac{-4 \pm \sqrt{16-4}}{2}$

$x = \frac{6 \pm \sqrt{36-4}}{2}$

$= -2 \pm \sqrt{5}$

$= 3 \pm 2\sqrt{5}$

(iv). $\tan^{-1}(\ln x)$

$f(u) = \ln x$

$f'(u) = \frac{1}{x}$

$\frac{d}{dx} (\tan^{-1}(\ln x)) = \frac{1}{x(1+(\ln x)^2)}$

Q1 (cont)

(v). $(3a-2b)^4$

$T_3 = T_2 + 1$

$= \binom{4}{2} (3a)^2 \cdot (2b)^2$

$= 6 \times 9a^2 \times 4b^2$

$= 216a^2b^2$

Question 2.

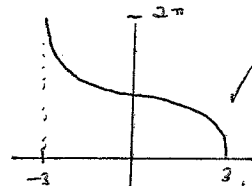
(i). $y = 2 \cos^{-1}(x/3)$

$\theta: -1 \leq x/3 \leq 1$

$-3 \leq x \leq 3$

$\theta: 0 \leq y/2 \leq \pi$

$0 \leq y \leq 2\pi$



(ii) $3 \sin \theta + 4 \cos \theta = 5$

$5 \sin(\theta + 53^\circ) = 5$

$\sin(\theta + 53^\circ) = 1$

$\theta + 53^\circ = 90^\circ$

$\theta = 37^\circ$

Note: $0 \leq \theta \leq 2\pi$

$\theta =$

(iii). $P(6,9)$

$A(1,4)$ $B(x,y)$

$x = 6 = \frac{m x_1 + n x_2}{m+n}$

$y = 9 = \frac{m y_1 + n y_2}{m+n}$

$6 = \frac{-3x + 2}{-1}$

$9 = \frac{-3y + 8}{-1}$

$-6 = -3x + 2$

$-9 = -3y + 8$

$-8 = -3x$

$-17 = -3y$

$x = 8/3$

$y = 17/3$

(iv) (a) Let $\angle RST = \alpha$

$\angle QRS = \alpha$ (Angles in the alternate segment)

$\angle QRS = \alpha$ (Angles in ||)

$\therefore \angle SQR = \angle SRQ = \alpha$

$\therefore \angle QRS = \alpha$ (Angles are equal)

(b) $\angle RST = \angle SPR = \alpha$ (Angles in the alternate segment)

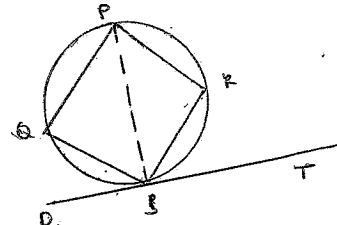
Since $\angle QRS = \alpha$

then $\angle QSR = \alpha$ (Angles in the alternate segment)

then $\angle QPS = \angle QSR$ (Angles in ||)

$\therefore \angle QPS = \angle SPR = \alpha$

$\therefore$ SP bisects $\angle QPR$ (Angles are equal)



Question 3.

(i). $P = \frac{1}{10}$ $Q = \frac{1}{10}$
 Hit $N = 11$ Hit

$(\frac{1}{10} + \frac{1}{10})^{20}$

$T_3 = \binom{20}{3} (\frac{1}{10})^3 (\frac{9}{10})^{17}$

$$(i) 2x^3 - 3x^2 + 0.99 = 0.$$

$$p(0) = 0.99$$

$$p(1) = -0.01$$

Change is sign, means root between 0 & 1.
ie near 1.

$$(b). x_0 = 1$$

$$p(x) = 6x^2 - 6x$$

$$x_1 = 1 - \frac{p(1)}{p'(1)}$$

$$= 1 - \frac{0.99}{0}$$

Cannot find x_1 .

(c). Fails because of turning point at $x=1$, ie $p'(x)=0$.

$$(iii). 2x^3 + 3x^2 + ax - 6.$$

$$p(-3) = 0.$$

$$2(-3)^3 + 3(-3)^2 + a(-3) - 6 = 0.$$

$$-33 - 3a = 0.$$

$$-3a = 33$$

$$a = -11$$

$$\text{Factors are } (x+3) \text{ and } (2x+3)$$

$$= 2x^2 + (4+6)x + 3 \times 3$$

$$2x^2 + 10x + 9 = 0$$

$$2x^2 + 3x + 3x + 9 = 0$$

$$x(2x+3) + 3(x+3) = 0$$

$$(2x+3)(x+3) = 0$$

$$\text{Let } p = -3$$

$$\text{Let } \beta = -3$$

$$\sum x = -3/2 = -3 + \beta$$

$$3/2 = \beta$$

$$\therefore \beta = -3$$

Question 4.

$$(i). \sum_{k=1}^n 5^k = 5 \left(\frac{5^n - 1}{4} \right)$$

$$\text{Let } n=1$$

$$\text{LHS} = 5^1 = 5$$

$$\text{RHS} = 5$$

$$\therefore \text{LHS} = \text{RHS}$$

True for $n=1$

Assume true for $n=k$

$$5^1 + 5^2 + \dots + 5^k = 5 \left(\frac{5^k - 1}{4} \right)$$

Prove true for $n=k+1$

$$5^1 + 5^2 + \dots + 5^k + 5^{k+1} = 5 \left(\frac{5^{k+1} - 1}{4} \right)$$

$$\text{LHS} = 5 \left(\frac{5^k - 1}{4} \right) + 5^{k+1}$$

$$= 5 \left(\frac{5^k - 1}{4} \right) + 5 \times 5^k$$

$$= 5 \left(\frac{5^k - 1}{4} + 4 \times 5^k \right)$$

$$= 5 \left(\frac{5^k + 4 \times 5^k - 1}{4} \right)$$

$$= 5 \left(\frac{5^k(4+1) - 1}{4} \right)$$

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If true for $n=k$, and then true for $n=k+1$.
By the principles of Mathematical Induction true for all n integers.

Q4 (cont)

$$(ii) (a) \int e^{f(x)} \cdot f'(x)$$

$$= \frac{e^{f(x)} + c}{f'(x)}$$

$$(b) \int_0^1 \frac{\cos^{-1} x}{\sqrt{1-x^2}} dx$$

$$= \int_0^1 \frac{-e^{\cos^{-1} x}}{\sqrt{1-x^2}} dx$$

$$= - \left[e^{\cos^{-1} x} \right]_0^1$$

$$= \left[e^{\cos^{-1} x} \right]_1^0$$

$$= e^{\pi/2} - e^0 = e^{\pi/2} - 1$$

$$(iii) (a) p(2ap, ap^2) \text{ and } (2aq, aq^2)$$

$$x^2 = 4ay$$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

$$= \frac{x}{2a}$$

$$\text{at } x = 2ap$$

$$m = p$$

$$\therefore \text{Eqn of } 1 = y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$

$$(b). \text{Eqn of } p = \frac{aq^2 - ap^2}{2aq - 2ap}$$

$$= \frac{q - p(q+p)}{2(q-p)}$$

$$= \frac{p+q}{2}$$

$$\text{Eqn: } y - ap^2 = \frac{p+q}{2} (x - 2ap)$$

$$2y - 2ap^2 = (p+q)x - 2ap(p+q)$$

$$2y - 2ap^2 = (p+q)x - 2ap^2 - 2apq$$

$$y = \frac{(p+q)x}{2} - apq$$

If focal chord, then passes through $(0, a)$

$$a = -apq$$

$$\frac{-a}{apq} = 1$$

$$-1 = pq$$

$$(c). \tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$= \frac{p - \left(\frac{p+q}{2} \right)}{1 + \frac{p(p+q)}{2}}$$

$$= \frac{3 - 1.4}{1 + 1.2}$$

$$\theta = 17^\circ 6'$$

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Question 5.

(i) $\dot{v}^2 = 4x - x^2$

(a). Let $v=0$.

$$4x - x^2 = 0.$$

$$x = 4 \neq 0.$$

$$0 \leq x \leq 4. \checkmark$$

(b). Centre $x=2$. $\checkmark$

$$\frac{1}{2}v^2 = 2x - \frac{x^2}{2}$$

$$\frac{d}{dx} \left(\frac{1}{2}v^2 \right) = 2 - x.$$

$$\ddot{x} = -[x-2]$$

when $x=2=0$. $\checkmark$

$$\therefore x=2.$$

(c). Max Speed, when $\ddot{x}=0$.

$$\Rightarrow x=2$$

$$v^2 = 8 - 4.$$

$$v^2 = 4.$$

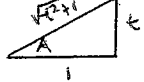
$$v = 2.$$

Max Speed is 2 m/s

(d). $\ddot{x} = -(x-2)$ $\checkmark$ (from above).

(ii). $\frac{2 \cos A}{\operatorname{cosec} A - 2 \sin A} = \tan 2A$

Let $\tan A = t$



$$\sin A = \frac{t}{\sqrt{t^2+1}}$$

$$\cos A = \frac{1}{\sqrt{t^2+1}}$$

$$\therefore \text{LHS} = \frac{2 \cos A}{\operatorname{cosec} A - 2 \sin A}$$

$$= \frac{2}{\frac{1}{\sqrt{t^2+1}} - \frac{2t}{\sqrt{t^2+1}}}$$

$$= \frac{2}{\frac{1-2t}{\sqrt{t^2+1}}}$$

$$= \frac{2 \sqrt{t^2+1}}{1-2t}$$

$$= \frac{2}{\frac{1-2t}{\sqrt{t^2+1}}}$$

$$= \frac{2t}{1-t^2}$$

$$= \frac{1+\tan A}{1-\tan A}$$

$$= \tan 2A$$

$$= \text{RHS}$$

$$\text{RHS} = \tan 2A$$

$$= \frac{2 \tan A}{1 - \tan^2 A}$$

$$= \frac{2t}{1-t^2}$$

Question 5.

(ii). $\frac{2 \cos A}{\operatorname{cosec} A - 2 \sin A} = \tan 2A$

Let $t = \tan A$



$$\sin A = \frac{t}{\sqrt{t^2+1}}$$

$$\cos A = \frac{1}{\sqrt{t^2+1}}$$

$$\text{LHS} = \frac{2t}{\sqrt{t^2+1}}$$

$$= \frac{2t}{\frac{1}{\sqrt{t^2+1}} - \frac{2t}{\sqrt{t^2+1}}}$$

$$= \frac{2}{\frac{1-2t}{\sqrt{t^2+1}}}$$

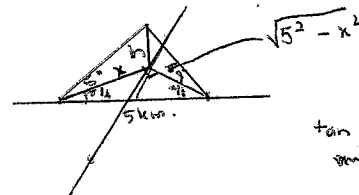
$$= \frac{2t}{1-t^2}$$

$$= \frac{2 \tan A}{1 - \tan^2 A} \checkmark$$

$$= \tan 2A$$

$$= \text{RHS}$$

(iii). (a)



$$\tan 5^\circ = \frac{h}{5}$$

$$\cot 5^\circ = \frac{5}{h}$$

$$h \cot 5^\circ = 5$$

$$h^2 \cot^2 5^\circ = 25$$

$$\tan 8^\circ = \frac{h}{\sqrt{5^2 + h^2}}$$

$$\cot 8^\circ = \frac{\sqrt{5^2 + h^2}}{h}$$

$$h \cot 8^\circ = \sqrt{5^2 + h^2}$$

$$h^2 \cot^2 8^\circ = 5^2 + h^2$$

$$5^2 - h^2 \cot^2 8^\circ = h^2$$

$$h^2 \cot^2 25^\circ = 25 - h^2 \cot^2 8^\circ$$

$$h^2 \cot^2 25^\circ + h^2 \cot^2 8^\circ = 25$$

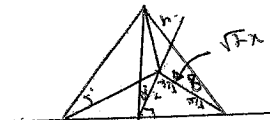
$$h^2 [\cot^2 25^\circ + \cot^2 8^\circ] = 25$$

$$h^2 = \frac{25}{\cot^2 25^\circ + \cot^2 8^\circ}$$

$$h = \frac{5}{\sqrt{130.646 + 50.628}}$$

$$= \frac{5}{13.46}$$

$$= 0.371 \text{ m}$$



(b)

$$\tan \alpha = \frac{h}{x}$$

$$\cot \alpha = \frac{x}{h}$$

$$\sqrt{2} h \cot \alpha = \sqrt{2} x$$

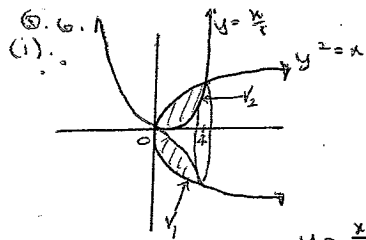
$$\tan 8^\circ = \frac{h}{\sqrt{2} x}$$

$$\cot 8^\circ = \frac{\sqrt{2} x}{h}$$

$$\sqrt{2} h \cot 8^\circ = \sqrt{2} x$$

$$\cot 8^\circ = \frac{\sqrt{2} x}{h}$$

$$\Rightarrow x = \tan^{-1} \left(\frac{\sqrt{2}}{7.115} \right)$$



$$y = \frac{x^2}{8}, \quad y^2 = x \Rightarrow y = \sqrt{x}$$

$$\left(\frac{x^2}{8}\right)^2 = x$$

$$\frac{x^4}{64} = x$$

$$x^4 - 64x = 0$$

$$x(x^3 - 64) = 0$$

$$x = 0, x = 4$$

$$\therefore \text{Area} = \int_0^4 \left(\sqrt{x} - \frac{x^2}{8} \right) dx$$

$$V_1 = \pi \int_0^4 (\sqrt{x})^2 dx$$

$$V_2 = \pi \int_0^4 \left(\frac{x^2}{8} \right)^2 dx$$

$$\therefore \text{Volume} = \pi \int_0^4 \left(x - \frac{x^{5/2}}{4} + \frac{x^4}{64} \right) dx$$

$$= \pi \left[\frac{x^2}{2} - \frac{2x^{7/2}}{28} + \frac{x^5}{320} \right]_0^4$$

$$= \pi \left[8 - 9^{1/4} + 3^{1/5} \right]$$

$$= 2^{2/5} \pi$$

(ii) (a) $f(x) = \sin^{-1} x + \cos^{-1} x = \pi/2$
 $f'(x) = \frac{d}{dx} \left(\pi/2 \right) = 0$

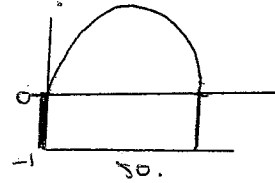
(b) $\int_0^{\pi/2} \sin^{-1} x + \cos^{-1} x \cdot dx$
 $= \int_0^{\pi/2} \pi/2 \cdot dx$
 $= \pi/2 \cdot x \Big|_0^{\pi/2}$
 $= \pi/2 \cdot \pi/2$

(iii) (a) $T = 20 + 50e^{-kt}$
 $\frac{dT}{dt} = -50ke^{-kt}$
 $= -k(T-20)$
 $T-20 = 50e^{-kt}$
 $\ln(T-20) = \ln 50 - kt$
 $\ln(40) = \ln 50 - 2k$
 $\ln(4/5) = -2k$
 $k = -\frac{\ln(4/5)}{2}$
 $\therefore T = 20 + 50e^{-\frac{\ln(4/5)}{2} t}$
 $= 20 + 50e^{-0.11t}$

(c) when $t = 4$
 $T = 20 + 50e^{-2 \ln 4/5}$
 $= 48 \frac{1}{8} ^\circ \text{C}$

(b) when $t = 2$, $T = 60^\circ \text{C}$
 $60 = 20 + 50e^{-2k}$
 $40 = 50e^{-2k}$
 $4/5 = e^{-2k}$
 $\ln(4/5) = -2k$
 $k = -\frac{\ln(4/5)}{2}$
 $\therefore T = 20 + 50e^{-\frac{\ln(4/5)}{2} t}$
 $= 20 + 50e^{-0.11t}$

Question 7.



(a) $x = vt \cos \theta, \quad y = -\frac{gt^2}{2} + vt \sin \theta$

$x = 50, t = 2, \quad y = 0, t = 2$
 $50 = 2v \cos \theta$
 $0 = -20 + 2v \sin \theta$
 $\frac{20}{\sin \theta} = 2v$

$$50 = \left(\frac{20}{\sin \theta} \right) \cos \theta$$

$$50 = 20 \cot \theta$$

$$50 \tan \theta = 20$$

$$\tan \theta = 2/5$$

$$\theta = \tan^{-1} (2/5)$$

$$= 21^\circ 48'$$

$$\frac{20}{\sin 21^\circ 48'} = 2v$$

$$v = \frac{20}{\sin 21^\circ 48'}$$

$$= 26.93 \text{ m/s}$$

(b) Max Height, when $t = 1$

$$y = -\frac{gt^2}{2} + vt \sin \theta$$

$$\text{sub } t = 1$$

$$y = -5 + 26.93 \sin 21^\circ 48'$$

$$= 5 \text{ m}$$

plus 1 metre, from start
 $\therefore \text{Max Height} = 6 \text{ m}$

(ii) $(a+bx)^n$

$$\frac{T_{r+1}}{T_r} = \frac{\binom{n}{r} (a+bx)^{n-r} b^r}{\binom{n}{r-1} (a+bx)^{n-r+1} b^{r-1}}$$

$$= \frac{n!}{(n-r)! r!} \times a^{n-r} \times (bx)^r$$

$$\frac{n!}{(n-r+1)! (r-1)!} \times a^{n-r+1} \times (bx)^{r-1}$$

$$= \frac{n!}{(n-r+1)! (r-1)!} \times \frac{(n-r+1)! (r-1)!}{n!} \times \frac{bx}{a}$$

$$= \frac{n-r+1}{r} \times \frac{bx}{a}$$

(b) $\frac{T_{r+1}}{T_r} = 1 \quad \therefore \frac{16-r}{r} \times \frac{1}{3} = 1$

$$\frac{16-r}{3r} = 1$$

$$16-r = 3r$$

$$16 = 4r$$

$$r = 4$$

$$\therefore T_5$$

$$\therefore T_5 \neq T_4$$